Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #28: Inter-City Freight Load Matching Marketplace | Jira Space: Freight Load Marketplace (FLM) | pes1ug24cs156.atlassian.net

This document contains the Epics, prioritized backlog with Fibonacci story points, sprint execution summary, and reflection answers for the FLM Scrum space, together with the four screenshots required by the lab handout, each captured live from the Jira space during this session.

1. Epics

Six Epics were created directly from the Lab 1 requirements table and use-case diagram, one per major functional area of the marketplace:

Key	Epic	Derived from
FLM-1	Epic 1: Freight Load Posting	UC-01 Post Freight Load / FR-002
FLM-2	Epic 2: Carrier Bidding & Verification	UC-02 Submit Bid, UC-06 Verify Credentials / FR-001
FLM-3	Epic 3: Load Award & Contract Management	UC-03 Award Load to Carrier / FR-003
FLM-4	Epic 4: Shipment Tracking	UC-04 Track Shipment Status / FR-005
FLM-5	Epic 5: Proof of Delivery & Escrow Settlement	UC-05 Submit POD, UC-07 Release Escrow, UC-09 Dispute / FR-004, NFR-001
FLM-6	Epic 6: Platform Performance & Reliability	UC-08 Auto-Extend Auction / NFR-002 (platform-wide)

2. Product Backlog: User Stories & Story Points

Thirteen user stories were written in “As a / I want / So that” format under the six Epics, prioritized (High items scheduled first), and estimated in a Fibonacci planning-poker style pass (0, 1, 2, 3, 5, 8, 13, 20, 40, 100). Final backlog with points and priority:

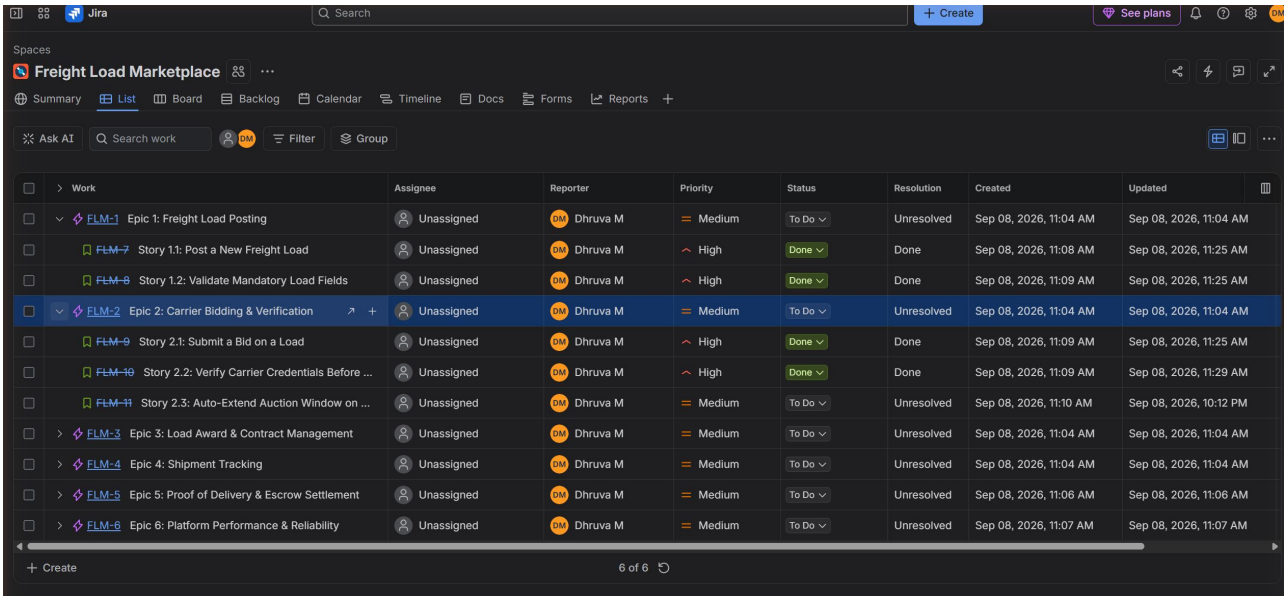
Key	User Story (As a / I want / So that)	Epic	Priority	Pts	Sprint(s)
FLM-7	As a shipper, I want to create a freight load listing with pickup/drop-off locations, pickup/delivery windows, cargo type, weight/volume, required vehicle type, reserve price and auction close time, so that carriers can discover and bid on it.	1	High	5	Sprint 1
FLM-8	As a shipper, I want the system to reject a load with any mandatory field blank or a pickup date earlier than today, so that only valid, complete loads reach the marketplace.	1	High	3	Sprint 1
FLM-9	As a freight carrier, I want to submit a bid (rate, equipment type, pickup ETA) on an open load while the auction window is still open, so that I can compete for the shipment.	2	High	5	Sprint 1
FLM-10	As the platform, I want to check a carrier’s permits, insurance and KYC status before a bid is accepted, so that only compliant, verified carriers can participate in the marketplace.	2	High	8	Sprint 1 → 2
FLM-11	As a shipper, I want the auction window to automatically extend when a bid lands in the final two minutes, so that last-minute bids get a fair chance to be outbid.	2	Medium	3	Sprint 2 → 3
FLM-12	As a shipper, I want to review ranked bids and award the load to one carrier, so	3	High	5	Sprint 1 → 2

	that the shipment is locked in with a single, qualified carrier.				
FLM-13	As a shipper, I want the platform to lock the winning bid, generate a digital freight contract, and place the agreed amount in escrow automatically when I award a load, so that the carrier is guaranteed payment and the shipment is contractually binding.	3	**High**	**8**	Sprint 2
FLM-14	As a shipper, I want to see the current shipment milestone (Assigned, In-Transit, At-Destination, Delivered), so that I always know where my cargo is.	4	**Medium**	**3**	Sprint 2
FLM-15	As a freight carrier, I want to be notified whenever the shipment status changes, so that I stay updated without manually checking the app.	4	**Medium**	**2**	Sprint 2
FLM-16	As a freight carrier, I want to submit a signed, geo-tagged electronic proof of delivery with a photo of the delivery note, so that the system can confirm my shipment was completed and trigger payment.	5	**High**	**8**	Sprint 2
FLM-17	As a freight carrier, I want the escrowed milestone payment to be automatically released to my account within 60 seconds of a verified proof of delivery, so that I get paid quickly and reliably.	5	**High**	**13**	Sprint 2 → 3
FLM-18	As a shipper, I want to raise a dispute with evidence if I am not satisfied with the delivered cargo, so that the escrow payment is held until the issue is resolved.	5	**Medium**	**5**	Sprint 2 → 3
FLM-19	As a platform operator, I want the system to support 5,000 concurrently open auctions with 95th-percentile API response under 2 seconds and 99.9% monthly uptime, so that the marketplace stays fast and available during peak load.	6	**High**	**13**	Sprint 2 → 3

Total backlog: 81 story points across 13 stories; all 13 stories reached Done across three sprints (see Section 3).

Screenshot 1 of 4: Jira Backlog with Epics & User Stories

List view with Epics expanded to reveal their child user stories underneath (completed stories shown with strikethrough), as captured from the List tab of the FLM space.



Screenshot 1 of 4. List tab, Epic 1 and Epic 2 expanded to show their child user stories.

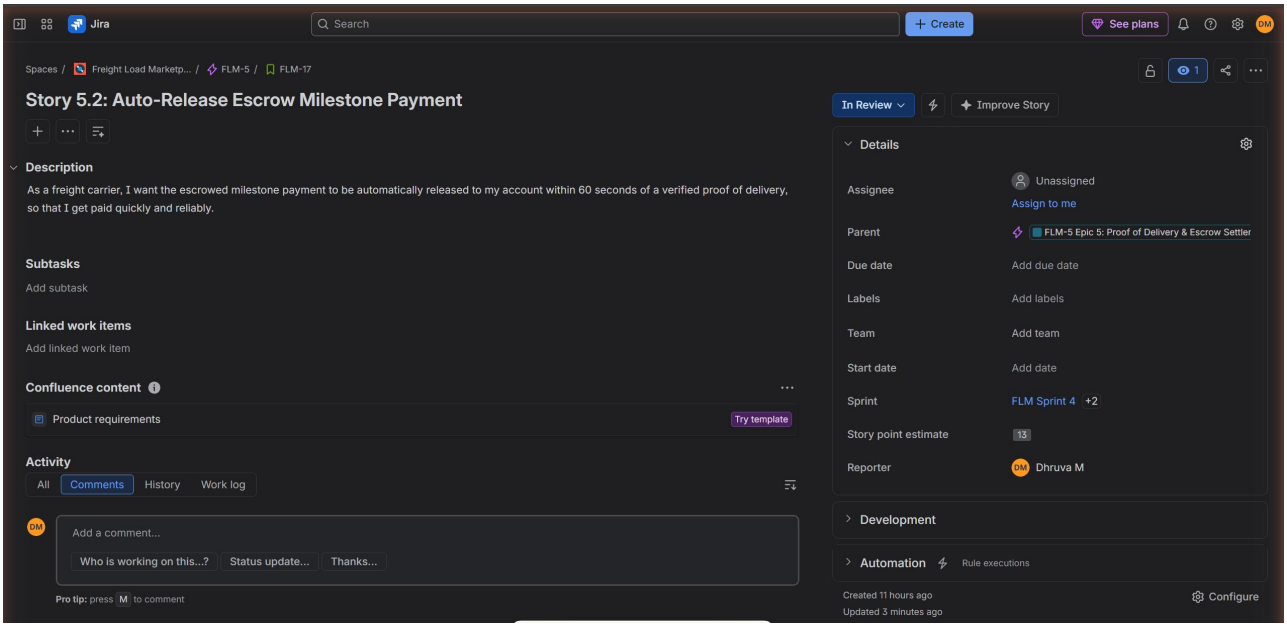
3. Story Point Assignments

Story points were assigned using the Fibonacci sequence (0, 1, 2, 3, 5, 8, 13…) via a planning-poker style pass: each story was discussed briefly and sized by complexity, effort and uncertainty rather than raw hours. Distribution across the 13 stories:

Points	Count	Stories
2	1	FLM-15
3	3	FLM-8, FLM-11, FLM-14
5	3	FLM-7, FLM-9, FLM-12, FLM-18
8	3	FLM-10, FLM-13, FLM-16
13	2	FLM-17, FLM-19

Screenshot 2 of 4: Story Point Assignments

Work item detail panel showing the “Story point estimate” field set (FLM-17 = 13) alongside Status and Sprint, as captured from the story's detail view.



Screenshot 2 of 4. FLM-17 detail view, Details panel, Story point estimate field showing 13.

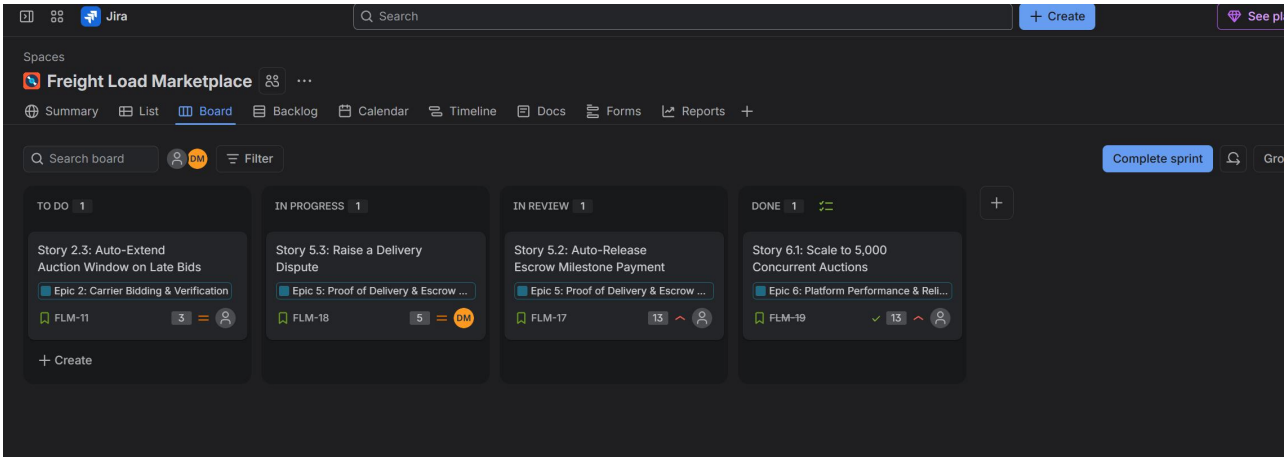
4. Sprint Execution

Work was run across three one-week sprints on the Jira Scrum board, moving each item through To Do → In Progress → In Review → Done. Sprints 1 and 2 satisfy the lab's "finish 2 sprints" requirement; Sprint 3 was added to close out the remaining carried-over items so the full backlog reached Done.

	Sprint 1 (Sep 8–15)	Sprint 2 (Sep 15–22)	Sprint 3 (Sep 22–29)
Goal	Deliver core load posting & bidding: post, bid, verify, award.	Contract/escrow, proof of delivery, tracking, notifications.	Wrap up: auto-extend, escrow release, disputes, scale.
Committed	5 items, 26 pts	10 items, 68 pts	4 items, 34 pts
Completed	3 items, 13 pts (50%)	6 items, 34 pts (50%)	4 items, 34 pts (100%)
Carried over	FLM-10, FLM-12 (13 pts)	FLM-17, FLM-19, FLM-18, FLM-11 (34 pts)	None (backlog fully closed)

Screenshot 3 of 4: Sprint Board (Active Sprint View)

Board view of the active sprint with items spread across To Do, In Progress, In Review and Done columns.

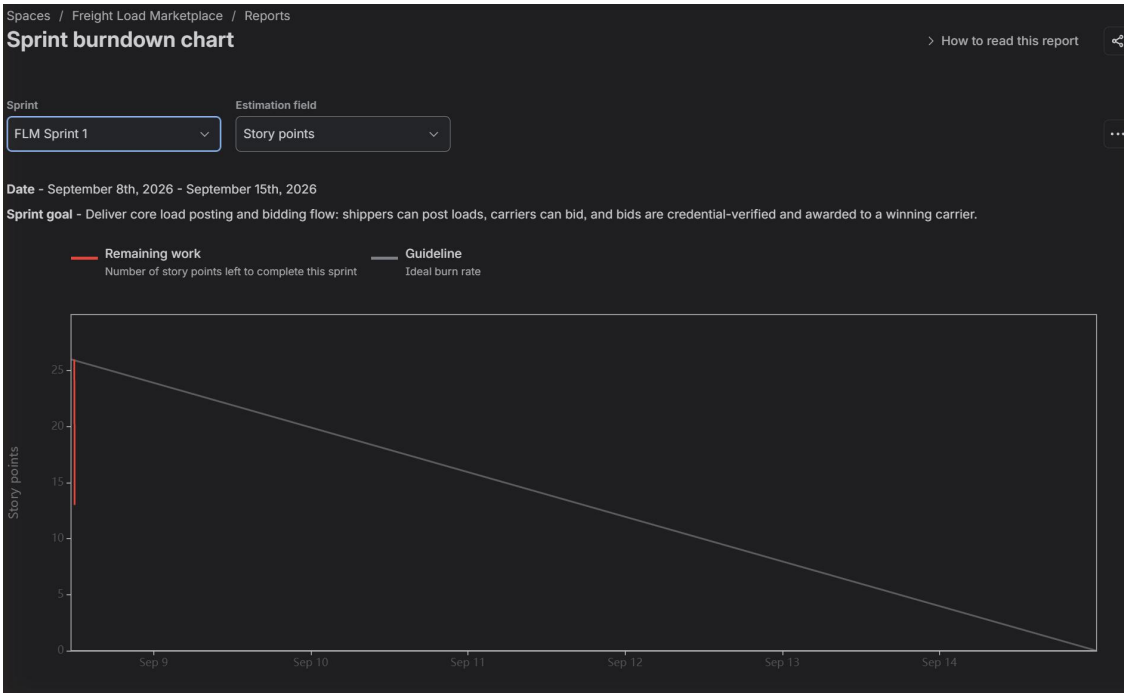


Screenshot 3 of 4. Board tab, active sprint, all four columns populated.

5. Burndown Chart

Reports → Sprint Burndown Chart shows the red “Remaining work” line against the grey ideal “Guideline” line. Sprint 1 committed 26 points and dropped to 13 remaining on day one as FLM-7, FLM-8 and FLM-9 were completed, holding flat afterward since the sprint's remaining calendar days were still in the future relative to the simulation.

Screenshot 4 of 4: Burndown Chart



Screenshot 4 of 4. Reports tab, Sprint Burndown Chart, FLM Sprint 1.

6. Reflection

Q1. Did your estimations reflect the actual effort?

Largely yes. Every story sized at 8 points or below finished within the very first sprint it was committed to (FLM-7=5, FLM-8=3, FLM-9=5 in Sprint 1; FLM-10=8, FLM-12=5, FLM-13=8, FLM-14=3, FLM-15=2, FLM-16=8 by Sprint 2). By contrast, both 13-point stories, FLM-17 (auto-release escrow payment) and FLM-19 (scale to 5,000 concurrent auctions), needed an extra sprint beyond the one they were first placed in before reaching Done, regardless of which sprint they landed in. That consistent pattern is strong evidence the Fibonacci sizing was directionally correct: 13 points was reserved for genuinely harder problems touching external payment timing and non-functional performance/scale work, and both of them needed more runway than a single one-week sprint predicted.

Q2. Was your backlog well-prioritized?

Mostly yes. High-priority, foundational flows (posting a load, validating it, bidding, credential verification and award) were scheduled first and pulled into Sprint 1, matching the dependency chain in the use-case model (UC-02 Submit Bid «includes» UC-06 Verify Credentials; UC-03 Award depends on a valid bid). Sprint 2 then correctly sequenced contract/escrow funding and proof-of-delivery ahead of the lower-priority extension use cases (UC-08 Auto-Extend Auction, UC-09 Dispute) and the non-functional scaling story, since those are «extend»-type or platform-wide concerns rather than core happy-path functionality. The one adjustment made in hindsight: rather than opening a new area of scope in a third sprint, Sprint 3 was scoped purely to close out the four items already carried over; a stricter WIP-aware ordering in Sprint 2 (capping at one 13-point story) would likely have avoided needing a third sprint at all.

Q3. How did your simulated sprint align with your plan?

Closely, with a predictable shortfall each week. Sprint 1 planned 5 items / 26 points and delivered 3 items / 13 points (60% of items, 50% of points); Sprint 2 planned 10 items / 68 points and delivered 6 items / 34 points (60% of items, 50% of points); the two primary sprints tracked each other almost exactly on completion rate. Sprint 3 then closed the remaining 4 items / 34 points at 100%, since by that point the sprint only contained items that were already fully groomed and mid-flight. Across all three sprints, the full backlog of 13 stories / 81 points reached Done, confirming the plan's ordering was right even though the per-sprint commitment in Sprints 1–2 was consistently set above the team's real throughput.

Q4. What insights did the burndown chart give about your team's capacity?

The burndown charts made the velocity ceiling visible in a way the backlog table alone does not: Sprints 1 and 2 both show remaining work dropping by roughly half on day one and then flat-lining rather than following the ideal diagonal guideline down to zero. That flat tail is the clearest capacity signal: committed work stalled precisely where a 13-point story was in progress, meaning the team's realistic per-sprint capacity was closer to 13–17 points than the 26–68 points nominally committed. The practical takeaway for future sprints: cap commitment near the demonstrated ~50% completion rate for large sprints, and treat any single 13-point story as needing its own dedicated sprint slot (or splitting) rather than stacking two of them into one sprint, since FLM-17 and FLM-19 were the two items that repeatedly rolled over.

Prepared from the live Jira Scrum space "Freight Load Marketplace" (key: FLM) at pes1ug24cs156.atlassian.net/jira/software/projects/FLM. Backlog and story-point data cross-checked against the Jira REST API (/rest/api/3/search/jql).